

Stage 1: The Conference Paper (Validation + Baseline)

Title: *Validating AI Perception Scales and Establishing Baseline Attitudes Among Malaysian Physics Students*

Conference Abstract:

The increasing adoption of artificial intelligence (AI) in education necessitates valid instruments to measure students' psychological perceptions, particularly within STEM contexts. This study evaluates the psychometric properties of adapted versions of the Short GAAIS-10 and AI-EdTech Trust Scale among Malaysian matriculation Physics students and establishes baseline attitudes and trust levels. A cross-sectional survey was conducted with 88 students (58% female, 68% from Life Sciences) from Sarawak Matriculation College. Data were analyzed using Jamovi for reliability (Cronbach's α and McDonald's ω) and descriptive statistics. All instruments demonstrated excellent internal consistency, with α/ω values of 0.902/0.903 for Positive Attitudes, 0.825/0.828 for Negative Dependency Concerns, and 0.921/0.922 for AI-EdTech Trust. Descriptively, students reported high positive attitudes towards AI ($M = 4.08$, $SD = 0.69$) and high trust in AI for supporting Physics learning ($M = 3.92$, $SD = 0.69$), alongside moderate negative dependency concerns ($M = 3.23$, $SD = 0.90$). The majority of students reported using AI "Often" (56.8%), indicating high digital engagement. These findings confirm that the adapted instruments are reliable for assessing AI perceptions in Malaysian educational contexts. The results establish a robust baseline, indicating that Malaysian matriculation Physics students are digitally confident and receptive to AI integration, providing a foundation for more complex psychological investigations.

Keywords: instrument validation, AI attitudes, trust in AI, STEM education, Malaysian students